

CSCI-3171 Network Computing

Assignment 1 – Winter 2026

Submission Instructions:

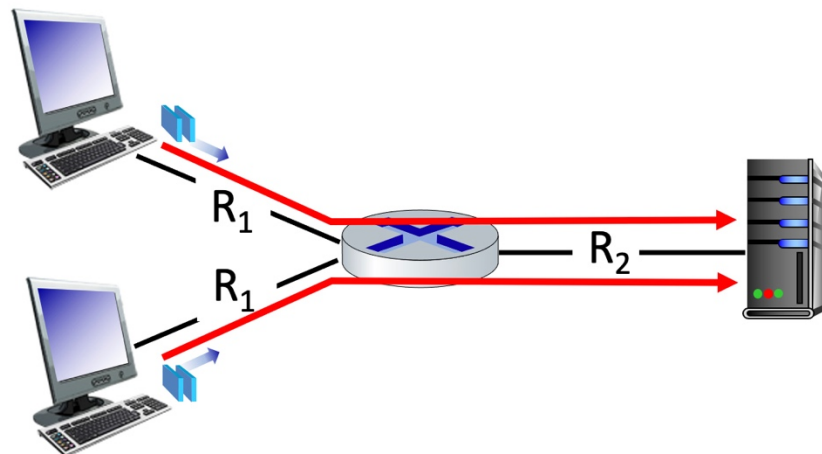
Submit **one PDF file** for your assignment on Brightspace under the Assignment 1 portal. Please ensure the following components are included:

1. Provide comprehensive steps for any computations, along with explanations of your methodology. If you made assumptions, clearly explain and justify them. Just writing the final answer will result in points being deducted.
2. Ensure that your full name and banner ID are clearly mentioned in the report.

Late submission: Refer to the syllabus on Brightspace for a complete description of the late submission policy.

Problem 1

Consider the network shown in the below, with two senders on the left sending packets to a common receiver on the right. The links have transmission rates of $R_1 = R_2 = 100$ Mbps (i.e., 100×10^6 bits per second). Suppose each packet is 1 Mbit (10^6 bits) in size.



- Q1.** How long does it take a sender to transmit a packet into its link (i.e., the packet transmission delay at the sender)? (1 point calculation, 1 point answer)

Assume that the propagation delay is 1 msec per link.

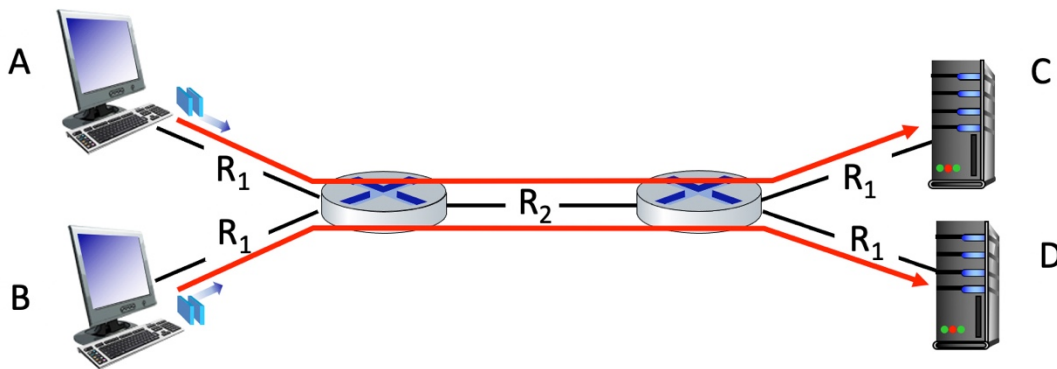
- Q2.** What is the end-to-end delay of a packet from when it first begins transmission at the sender, until it is received in full by the server at the end of the rightmost link. Assume store-and-forward packet transmission. You can ignore the queuing and processing delays. (1 point calculation, 1 point answer)

Assume that both sessions are sending packets simultaneously at the maximum rate possible and that the link R_2 is equally shared between the two sessions.

- Q3.** What is the maximum end-to-end throughput achieved by each session in this case? (1 point)
Q4. What is the obtained utilization ratio of the link connecting the sender to the router? (1 point)

Problem 2

Consider the network shown below, with two senders (A,B) on the left, sending packets to two different receivers (C,D) on the right. Each sender is sending packets to its receivers over a separate connection. The links have transmission rates of $R_1 = 100$ Mbps (i.e., 100×10^6 bits per second) and $R_2 = 150$ Mbps. Suppose each packet is 1 Mbit (10^6 bits) in size.



- Q5.** What is the packet transmission delay at sender A? (1 point calculation, 1 point answer)

Assume that the propagation delay is 10 msec per link. You can ignore the queuing and processing delays. Note that each of the two sessions has a 3-hop path from source to destination.

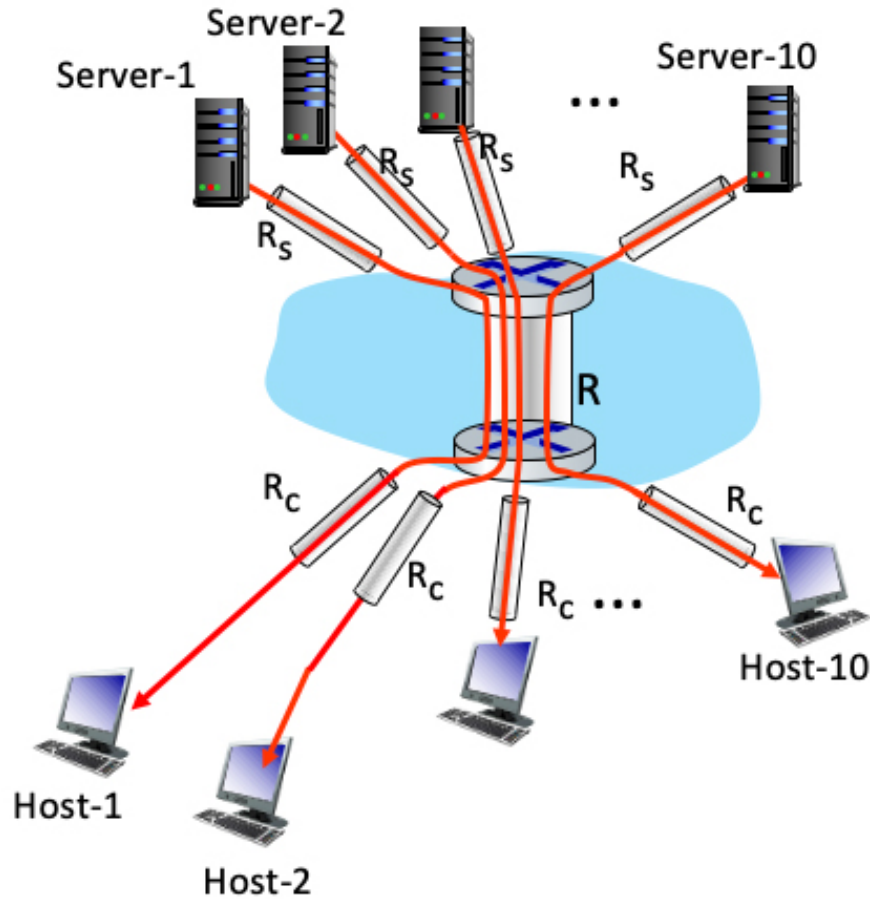
- Q6.** What is the end-to-end delay of a packet from when it first begins transmission at a sender, until it is received in full by a receiver at the right. Assume store-and forward packet transmission. (1 point calculation, 1 point answer)

Assume that both sessions are sending packets simultaneously at the maximum rate possible and that the link with capacity R_2 is equally shared between the two sessions.

- Q7.** What is the maximum end-to-end throughput achieved by each session in this case? (1 point)
Q8. What is the obtained utilization ratio of the shared link with capacity R_2 ? (1 point)

Problem 3

Consider the scenario shown below, with 10 different servers (four shown) connected to 10 different clients over ten three-hop paths. The pairs share a common middle hop with a transmission capacity of $R = 200$ Mbps. Each link from a server to the shared link has a transmission capacity of $R_s = 25$ Mbps. Each link from the shared middle link to a client has a transmission capacity of $R_c = 50$ Mbps.

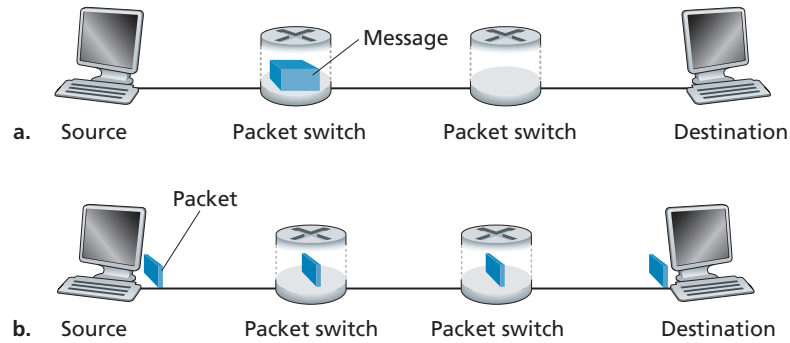


Now consider the three-hop path from Server-10 to host-10, assuming the values of R_s , R_c , and R given above. Assume also that a packet is 1000 bits long, and that each link has a propagation delay of 100 microseconds (1 microsec = 0.000001 sec). You can assume that queueing delay and processing delays are zero.

- Q9.** What is the total amount of time from when a server starts sending a packet until a host completely receives that packet? (4 points calculation, 1 point answer)
- Q10.** What is the maximum achievable end-end throughput (in Mbps) for each of the ten client-to-server pairs, assuming that the middle link is equally shared and all servers are trying to send at their maximum rate? (0.5 point calculation, 0.5 point answer)

Problem 4

In modern packet-switched networks, including the Internet, the source host segments long, application-layer messages (for example, an image or a video file) into smaller packets and sends the packets into the network. The receiver then reassembles the packets back into the original message. We refer to this process as *message segmentation*. The figure below illustrates the end-to-end transport of a message with and without message segmentation. Consider a message that is 8×10^6 bits long that is to be sent from source to destination. Suppose each link in the figure is 2 Mbps. Ignore propagation, queuing, and processing delays.



- Q11.** Consider sending the message from source to destination without message segmentation. How long does it take to move the message from the source host to the first packet switch? Keeping in mind that each switch uses store-and-forward packet switching, what is the total time to move the message from source host to destination host? (2 point calculation, 1 point answer)
- Q12.** Now suppose that the message is segmented into 800 packets, with each packet being 10,000 bits long. How long does it take to move the first packet from source host to the first switch? At what time will the second packet be fully received at the first switch? And the third packet? (1 point calculation, 2 points answer)
- Q13.** How long does it take to move the file from source host to destination host when message segmentation is used? Compare this result with the case where no segmentation is used and comment. (2 point calculation, 1 point answer, 1 point comparison)
- Q14.** In addition to reducing delay, what are reasons to use message segmentation? (1 point)
- Q15.** Discuss the drawbacks of message segmentation. (1 point)